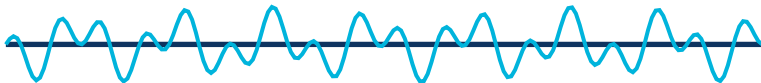


Applications of DFT



Spectral Analysis of Signals

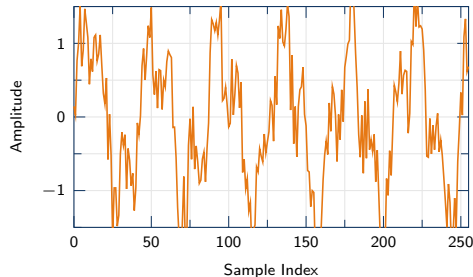
Observation:

- Start by simply **looking at the data**
- 256 samples from the experiment
- Human eye sees only a **noisy waveform**
- Very little useful information is visible

Key Problem

256 points provide **insufficient information** for a well-behaved frequency spectrum after DFT. The result is a noisy mess.

Raw Signal (256 pts)



Spectral Analysis of Signals

The Spectral Leakage Problem:

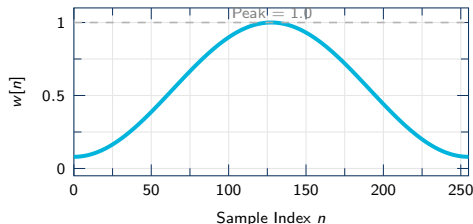
- DFT assumes the signal is **periodic** and repeats infinitely
- A finite segment creates **discontinuities** at its edges
- These edges introduce **spectral leakage** — energy from one frequency "leaks" into neighboring frequencies
- Result: distorted, smeared spectrum

The Fix — Hamming Window:

- Smoothly **tapers samples to zero** at both ends
- Eliminates artificial edge discontinuities
- Significantly reduces spectral leakage

$$w[n] = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right)$$

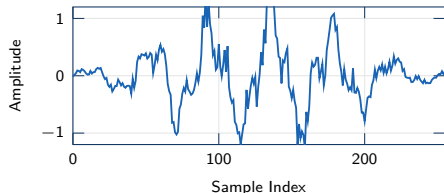
Hamming Window ($N = 256$)



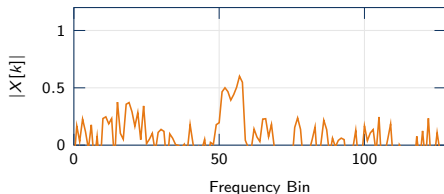
Effect: Samples at $n = 0$ and $n = 255$ are multiplied by ≈ 0.08 , while the center sample ($n = 128$) is multiplied by 1.0. This smooth bell-curve shape prevents abrupt edges.

Spectral Analysis of Signals

Signal $\times$ Hamming Window



DFT Magnitude Spectrum (129 pts)



DFT Result is Still Noisy

Even after windowing, a single 256-pt DFT yields a **noisy, hard-to-interpret** spectrum.

Why longer DFTs don't help:

- A 2048-pt DFT $\Rightarrow$ 1025-sample spectrum
- More data, but **information diluted** by same factor
- Better **frequency resolution**, but **same noise level**

Core Insight

We need to use more data **without** increasing spectrum length.

Solution: Segment averaging!

Spectral Analysis of Signals

The Averaging Process:

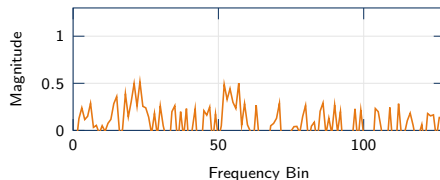
- 1 Break the long input signal into many **256-sample segments**
- 2 Multiply each segment by the **Hamming window**
- 3 Run each through a **256-point DFT**
- 4 Convert to polar notation
- 5 **Average the magnitude spectra** (phase discarded)

Noise Reduction Law

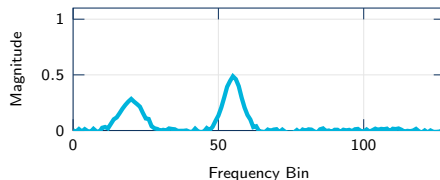
$$\text{Noise} \propto \frac{1}{\sqrt{N_{\text{segments}}}}$$

Averaging **100 segments** reduces noise by a factor of 10.

Single Segment Spectrum (noisy)



After Averaging 100 Segments (smooth)



Spectral Analysis of Signals

Alternative Approach:

- 1 Take a **very long DFT** (e.g. 16,384 points)
- 2 Spectrum is high-resolution (8193 bins) but **very noisy**
- 3 Apply a **low-pass digital filter** to smooth the spectrum
- 4 Trade resolution for reduced noise

Equivalence Example

Averaging 64 adjacent bins from a 16,384-pt DFT $\approx$ averaging 64 segments of 256 pts each.

Ultimate Strategy:

10,000 segments $\times$ 16,384 samples $\Rightarrow$

High resolution + Low noise. Problem solved!

Recommendation

Use Method 1 (segment averaging) in most cases. More data always improves both noise and resolution — collect more segments rather than adding a filter.

Anatomy of a Frequency Spectrum

A real-world spectrum has five recognizable regions

- **1/f noise** — rapid rise below ~ 10 Hz
- **White noise** — flat region, 10–70 Hz
- **Filter roll-off** — rapid drop above 70 Hz
- **Interference peaks** — 60 Hz and harmonics, RF, microphonics
- **Signal peaks** — the actual periodic signal of interest

Example: Fig. 9-2

Undersea microphone spectrum showing all five features simultaneously, including a 13 Hz signal buried among noise and interference.

White Noise (10–70 Hz)

Definition

A flat spectral region containing **equal amounts of all frequencies** — the frequency-domain analog of white light.

Time-domain origin

- Results when noise samples are **uncorrelated** sample-to-sample
- Knowing one noise value gives *no information* about any other sample

Sources

- Random motion of electrons in electronic circuits
- Water spray hitting a shower floor (familiar acoustic analogy)
- In Fig. 9-2: could come from analog electronics *or* the ocean itself

Antialias Filter Roll-off (Above 70 Hz)

What happens

Above 70 Hz, the white noise amplitude **rapidly decreases** — this is the roll-off of the antialias filter.

Ideal filter

Passes everything below 80 Hz; blocks everything above — a perfectly sharp cutoff.

Real filter

Cutoff is gradual, never perfectly sharp — the roll-off you actually observe.

Diagnostic warning

If you **don't** see this gradual roll-off, suspect an **aliasing problem**.

1/f Noise (Below ~ 10 Hz)

Observation

Below about 10 Hz, noise amplitude **rapidly increases** — known as $1/f$ (“one-over-f”) noise.

The mystery

- Measured in extremely diverse systems: traffic density on freeways, transistor electronic noise
- Probably present in *all* systems, if measured low enough in frequency
- Cause is identifiable in specific systems — but **why it's universal remains unexplained**

Transition point

For common analog electronics and most physical systems, the white noise $\leftrightarrow$ $1/f$ noise transition occurs between **1 and 100 Hz**.

60 Hz peak

Electromagnetic interference from commercial electrical power lines.

Harmonics

- Smaller peaks appear at **multiples** of 60 Hz: 120, 180, 240 Hz, etc.
- Occur because the real power-line waveform is **not a perfect sinusoid**

Common culprits

Source	Frequency range
Switching power supplies	25–40 kHz
Nearby radio / TV stations	Megahertz range
Microphonics (components vibrating when shaken)	10–100 Hz

Note

Microphonics is a favorite explanation for unexplained low-frequency peaks: mechanical vibration coupling into the electronics.

The Actual Signal: Fundamental & Harmonics

Spectral pattern

- Strong peak at **13 Hz** — the **fundamental frequency**
- Weaker peak at **26 Hz** — the **2nd harmonic**
- Weaker peak at **39 Hz** — the **3rd harmonic**
- Higher multiples (52, 65, 78 Hz, ...) exist but are **buried in white noise**

This is a nonsinusoidal periodic waveform

A periodic but non-sinusoidal signal produces energy at its fundamental frequency *and* at integer multiples of it.

Real-world example: Passive Sonar

A 13 Hz signal could be generated by a submarine's three-bladed propeller turning at **4.33 revolutions/second**. Identifying vessels by their frequency/harmonic “fingerprint” is the basis of passive sonar.

Frequency Resolution: Two Limiting Factors

The question

How close can two spectral peaks be before they merge into one (Fig. 9-3)?

Two independent limits

- 1 **DFT length** — how finely the spectrum is sampled
- 2 **Signal length** — how long the input waveform actually is

Key distinction

These are *not* the same thing — the length of the DFT does not have to equal the length of the input signal.

Factor 1: Length of the DFT

Sample spacing in the spectrum

An N -point DFT produces $N/2 + 1$ samples, equally spaced between 0 and half the sampling frequency.

To separate two close peaks

The sample spacing must be **smaller than the distance** between the two peaks.

Example

- A **512-point** DFT is sufficient to separate the peaks in Fig. 9-3
- A **128-point** DFT is *not* — the peaks merge

Factor 2: Length of the Input Signal

The subtler limit

Imagine two sine waves with only a slight frequency difference. Over a short segment, the sum looks like a single sine wave.

Consequence

The **closer the frequencies**, the **longer the segment** must be to reveal that more than one frequency is present.

Sinusoids Between Basis Functions (Fig. 9-4a)

Setup

Signal = two sine waves:

- One frequency **matches** a basis function
- One frequency falls **between** two basis functions

Result

- Matching sine wave → clean **single point**
- Off-bin sine wave → peak with **tails extending far away**, since it cannot be represented by one sample

The Fix: Multiply by a Hamming Window (Fig. 9-4b)

Three effects of windowing

- ① Peaks are made to **look more alike** — good
- ② Tails are **greatly reduced** — good
- ③ Peaks become **wider**, reducing resolution — bad

The fundamental trade-off

Windows trade **resolution** (peak width) against **spectral leakage** (tail amplitude).

Effect on the Frequency Domain: Convolution

Multiplication $\leftrightarrow$ Convolution

Multiplying two time-domain signals corresponds to **convolving** their frequency spectra.

Applying this here

- Original spectrum = delta function (infinitesimally narrow)
- Windowed signal's spectrum = **the window's own spectrum**, shifted to the peak's location

Consequence

The shape of the window *directly determines* the shape of every peak in the final spectrum (Fig. 9-5).

Comparing Windows (Fig. 9-5)

Every window broadens the peak and adds side lobes

This degradation is unavoidable — it results from using only a portion of the time-domain signal.

The trade-off across windows

Window	Main lobe	Tails
Rectangular	Narrowest (good)	Largest (bad)
Hamming	Medium	Medium
Blackman	Widest (bad)	Lowest (good)

The Windowed Spectrum Is Continuous

Why

After windowing, the time-domain signal is **still infinitely long** (mostly zeros) $\Rightarrow$ its spectrum consists of $\infty/2 + 1$ samples between 0 and 0.5 — effectively a **continuous curve**.

Step 2 needed

We must select N points from the time-domain signal that:

- Contain **all** nonzero samples defined by the window
- May also include any number of the surrounding zeros

Choosing N : Sampling the Continuous Curve

Example

$N = 1024 \Rightarrow$ the continuous spectral curve is sampled **513 times** between 0 and 0.5.

Two regimes

- $N \gg$ window length $\Rightarrow$ samples close together, curve's peaks/valleys **preserved**
- $N =$ window length $\Rightarrow$ fewer samples, regular peaks/valleys turn into **irregular tails**

Each peak = a sampling of the Fig. 9-5a curve

Whether tails appear depends entirely on **where** the samples land relative to the curve's peaks and valleys.

Two cases

- Sine wave **matches** a basis function → samples fall exactly at the **valleys** → tails vanish
- Sine wave falls **between** basis functions → samples fall along the slopes → visible tail pattern

The Flat-Top Window: Why?

Problem: measuring amplitude accurately

Since the DFT spectrum is only **samples**, nothing guarantees a sample lands exactly on a peak's top — the nearest sample is usually slightly off-center, **underestimating** the true amplitude.

Solution

Use a window whose spectral peak has a **flat top** (Fig. 9-5d), guaranteeing one or more samples hit the correct peak value.

Cost

A very **broad main lobe** $\Rightarrow$ poor frequency resolution.

A Subtle Amplitude Rule

Remember when interpreting peak amplitude

- A **DC** value of 1 in the time domain $\rightarrow$ peak amplitude of **1** in the frequency domain
- A **sinusoid** of amplitude 1 in the time domain $\rightarrow$ peak amplitude of only **0.5** in the frequency domain

Why

Explained via the Inverse DFT synthesis relationship (see Chapter: *Synthesis, Calculating the Inverse DFT*).

Two Views of a System

- **Time domain:** systems are analyzed using **convolution**

$$x[n] * h[n] = y[n]$$

- **Frequency domain:** an equivalent analysis exists
 - Using the **Fourier Transform**, any input signal can be written as a group of cosine waves, each with its own amplitude and phase shift
 - The DFT represents the output signal the same way
- ⇒ Any linear system can be **completely described** by how it changes the amplitude and phase of cosine waves passing through it

Definition

This complete description is called the system's **frequency response**.

Impulse Response $\leftrightarrow$ Frequency Response

- Both the impulse response and the frequency response contain **complete information** about the system
- Therefore, there is a **one-to-one correspondence** between them
 - Given one, the other can always be calculated

A Foundation of Signal Processing

A system's frequency response is the **Fourier Transform** of its impulse response.

Notation convention:

- Impulse response $\rightarrow$ lower case: $h[n]$
- Frequency response $\rightarrow$ upper case: $H[f]$

The Convolution Theorem

Time Domain

$$x[n] * h[n] = y[n]$$

(convolution)

Frequency Domain

$$X[f] \times H[f] = Y[f]$$

(multiplication)

Key Idea

Convolution in the time domain corresponds to **multiplication** in the frequency domain.

Example: Uncovering a Filter's Behavior

- (a) The impulse response alone gives no hint of what the system does
 - Taking a **64-point DFT** of the impulse response produces the frequency response
 - (b) The system's function is now obvious:
 - Passes frequencies between **0.2 and 0.3**
 - Rejects all others
- ⇒ This is a **band-pass filter**

Note: The phase of the frequency response could also be examined, but it is more difficult to interpret and less interesting at this stage.

Improving Frequency Resolution

- The frequency response curve is **jagged** due to the low number of samples defining it
- **Solution:** pad the impulse response with zeros before taking the DFT

Example

Padding the impulse response to **512 samples** (c) results in a much higher-resolution frequency response (d).

- More zero-padding $\Rightarrow$ finer detail on the *same* underlying curve

How Much Resolution Can We Get?

- Answer: **infinitely high** — if willing to pad with an infinite number of zeros
- Nothing limits frequency resolution except the length of the DFT

Important Concept

Even though the impulse response is a **discrete** signal, the corresponding frequency response is **continuous**.

- An N -point DFT of the impulse response provides only $N/2 + 1$ samples of this continuous curve
- Since the input can contain any frequency between 0 and 0.5, the frequency response *must* be continuous over this range

- Consider an N -sample signal run through an N -point DFT $\rightarrow N/2 + 1$ frequency samples
- Recall: the DFT treats the time domain signal as **infinitely long and periodic**
 - The N points repeat from negative to positive infinity
- Now pad the time domain with an increasing number of zeros:
 - The **period** of the time domain signal grows longer
 - The **spacing** between frequency-domain samples grows finer

Add an infinite number of zeros to the time domain signal:

- ① Time domain now has an **infinitely long period** $\Rightarrow$ becomes **aperiodic**
- ② Frequency domain achieves **infinitesimal spacing** $\Rightarrow$ becomes **continuous**

Discrete Time Fourier Transform (DTFT)

The procedure that changes a discrete aperiodic signal into a frequency domain that is a continuous curve.

A system's frequency response is found by taking the DTFT of its impulse response.

	DTFT	DFT
Domain used in	Mathematical equations	Computer calculation
Time signal	Aperiodic, infinite	Finite, periodic
Frequency result	Continuous curve	Discrete samples
Computable?	No	Yes

Bottom Line

Since the DTFT cannot be computed exactly, the **DFT** is used to calculate a *sampling* of the true, continuous frequency response.

Why Leave the Time Domain?

The idea

Transform signal and impulse response into the frequency domain, **multiply**, then transform back: DFT $\rightarrow$ Multiply $\rightarrow$ Inverse DFT

This replaces one convolution with two DFTs, a multiplication, and an inverse DFT — yet the result is identical to standard convolution.

Reason 1: Difficulty

- **Deconvolution** (input from output & impulse response) is nearly impossible in the time domain
- In the frequency domain it becomes a simple **division**

Reason 2: Speed

- 512-sample kernel: only $\sim 1,000$ samples/sec via direct convolution
- Too slow for audio ($40\times$) or video ($10,000\times$)
- The **FFT** makes it hundreds of times faster

Multiplying Frequency Domain Signals

Polar form (the intuitive picture)

A cosine wave entering a system emerges as a cosine wave with a new amplitude and phase:

$$\begin{aligned}\text{Mag}_Y[f] &= \text{Mag}_X[f] \times \text{Mag}_H[f] \\ \text{Phase}_Y[f] &= \text{Phase}_X[f] + \text{Phase}_H[f]\end{aligned}$$

Rectangular form

Multiplying in rectangular notation introduces cross terms between real and imaginary parts — for example, a sine wave in can produce *both* cosine and sine components out. The rectangular equations are messier, but describe the same physical operation as the polar form above.

Understand it via polar form first; treat the rectangular equations as bookkeeping.

Deconvolution: Undoing Convolution

The problem

Given the output $Y[f]$ and the impulse response $X[f]$, find the input:

$$H[f] = \frac{Y[f]}{X[f]}$$

Polar form solution

$$\begin{aligned}\text{Mag}_H[f] &= \text{Mag}_Y[f] / \text{Mag}_X[f] \\ \text{Phase}_H[f] &= \text{Phase}_Y[f] - \text{Phase}_X[f]\end{aligned}$$

- Division is the inverse of multiplication: magnitudes divide, phases subtract
- In rectangular form the same operation again produces cross terms
- This is **why** deconvolution is tractable in the frequency domain but nearly hopeless in the time domain

Circular Convolution: The Hidden Trap

The length mismatch

Convolving an N -point signal with an M -point impulse response produces $N + M - 1$ points — but an N -point DFT $\rightarrow$ multiply $\rightarrow$ inverse DFT can only return N points.

Example

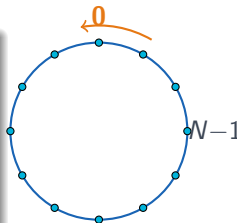
- Input: 256 points Impulse response: 51 nonzero points
- True convolution length: $256 + 51 - 1 = 306$ samples
- Frequency-domain method only provides 256 output points
- Result: 49 samples wrap around and **overlap** with the start — **circular convolution**

The DFT treats the N -point time domain as one period of an infinitely repeating signal — so the “extra” samples spill into the neighboring period and add back in.

Avoiding Circular Convolution

Why “circular”?

- The DFT's time domain behaves as if sample $N - 1$ sits next to sample 0
- A signal exiting the right edge reappears on the left — like a snake biting its own tail
- Distortion = one period's overflow bleeding into the next



Sample $N-1$ wraps to sample 0

The fix: zero-padding

Pad both signals with zeros so the DFT length $\geq N + M - 1$ (e.g. 256 and 51-point signals padded to 512). The output then contains the full 306 correct samples plus trailing zeros — no wraparound.